



Department of Computer Engineering
& Electronic Systems



INTERNATIONAL
HELLENIC
UNIVERSITY

Διαδίκτυο των Αντικειμένων

Internet of Things

LPWAN – RPL - Architectures

Μάριος Αγγελόπουλος



Evolution of Internet

Over the past decades, Internet has been growing at an **increasing rate**, incorporating more and more **types/classes of devices/systems**

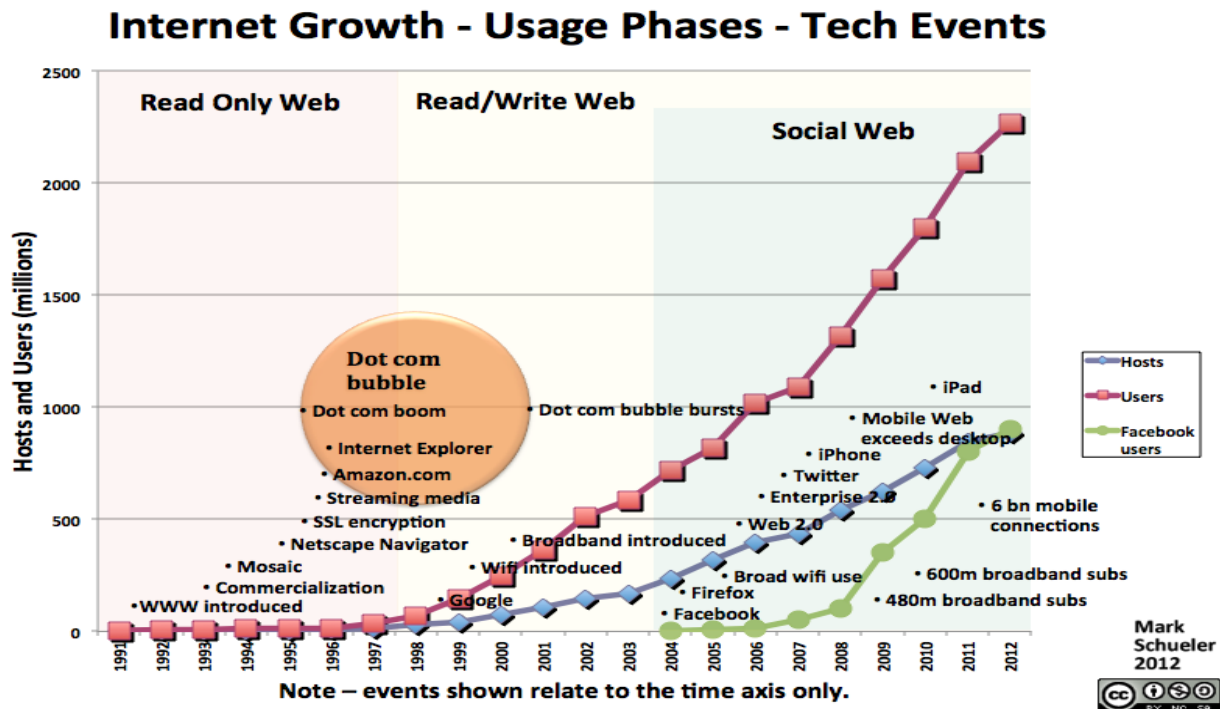
First wave: Computers => Telephony Nets => Satellites => Cellular Nets

Second wave: Humans (Social networks) – Dynamic content generation

Next wave: Machines / "Things" – Devices use the Internet infrastructure

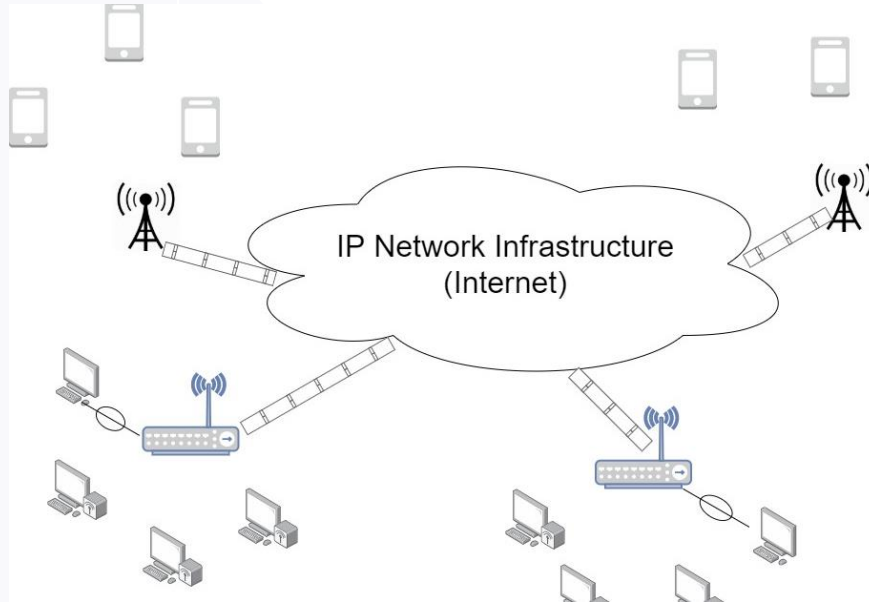


Evolution of Internet





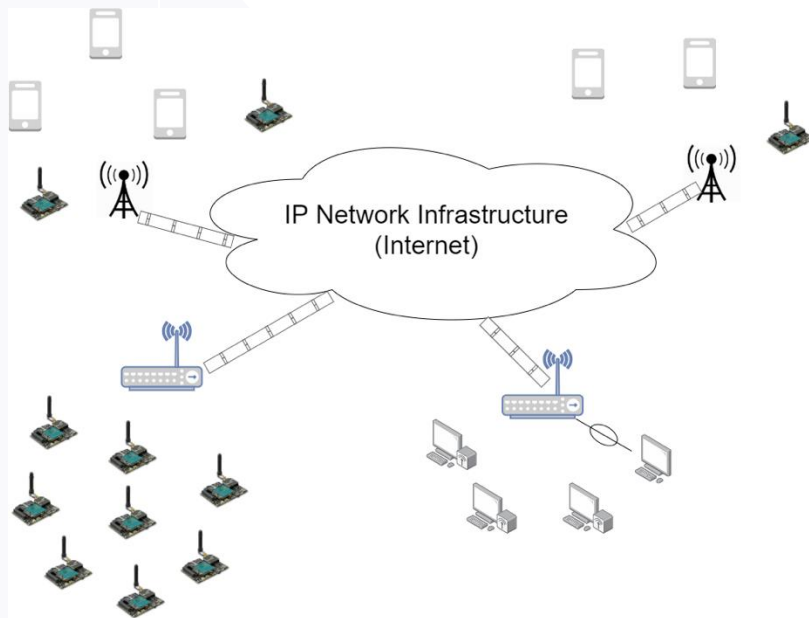
Typical Computer Networks



- End-nodes are powerful devices
- Connections are reliable and persistent
- More or less, static architectures or networks with low dynamics



IoT and Sensor Networks



- End-nodes are highly-constrained devices.
- Connections are unreliable and ephemeral.
- Significantly higher numbers of devices.
- New protocol stacks introduced – interoperability and standardization issues.



IoT Elements and Environments

Devices (tags, sensors, wearables, etc)



Machines (home appliances, robots, vehicles, etc)



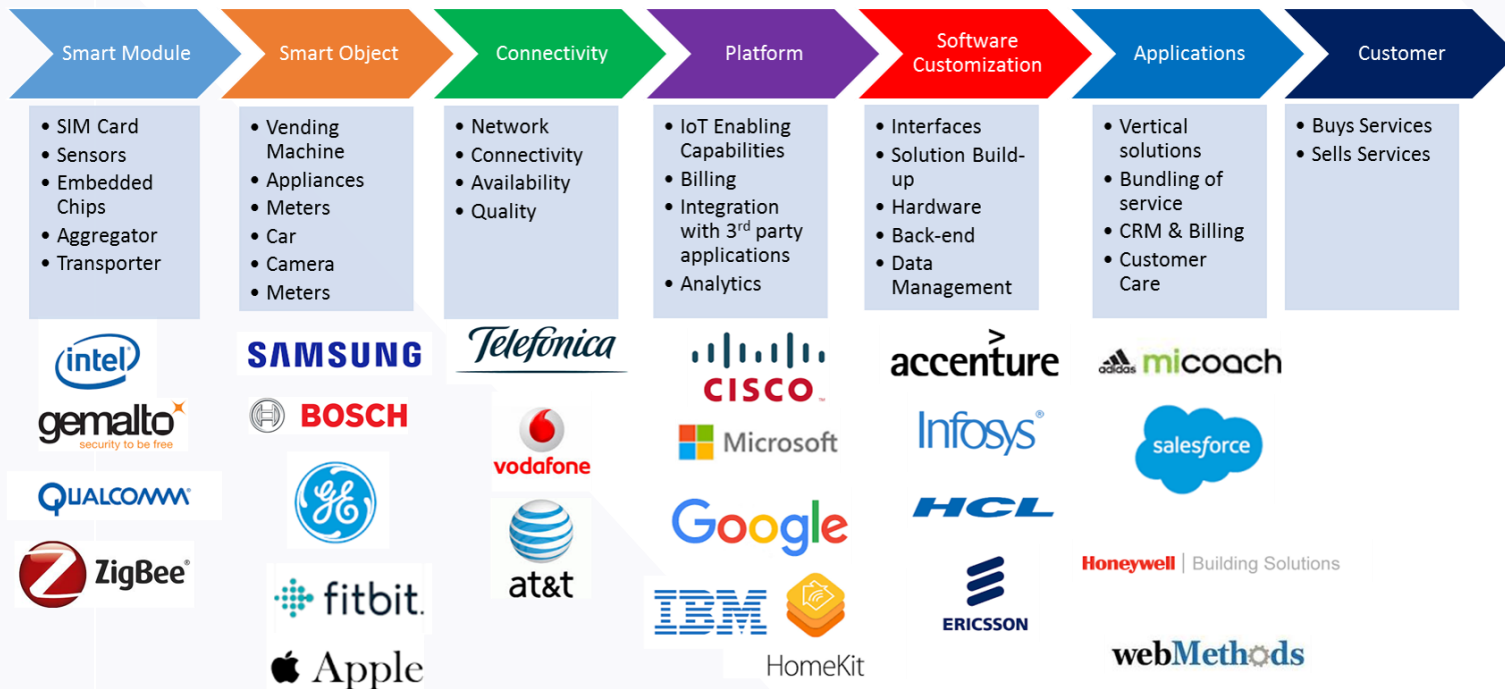
Environments (smart buildings, smart campus, smart city, smart economy, etc)





IoT Industrial Ecosystem

Internet of Things Value Chain



Note, the above is not an exhaustive list of companies and any company may have play in more than one component of value chain

Copyright: Telecomcircle.com



Wireless Communication Technologies

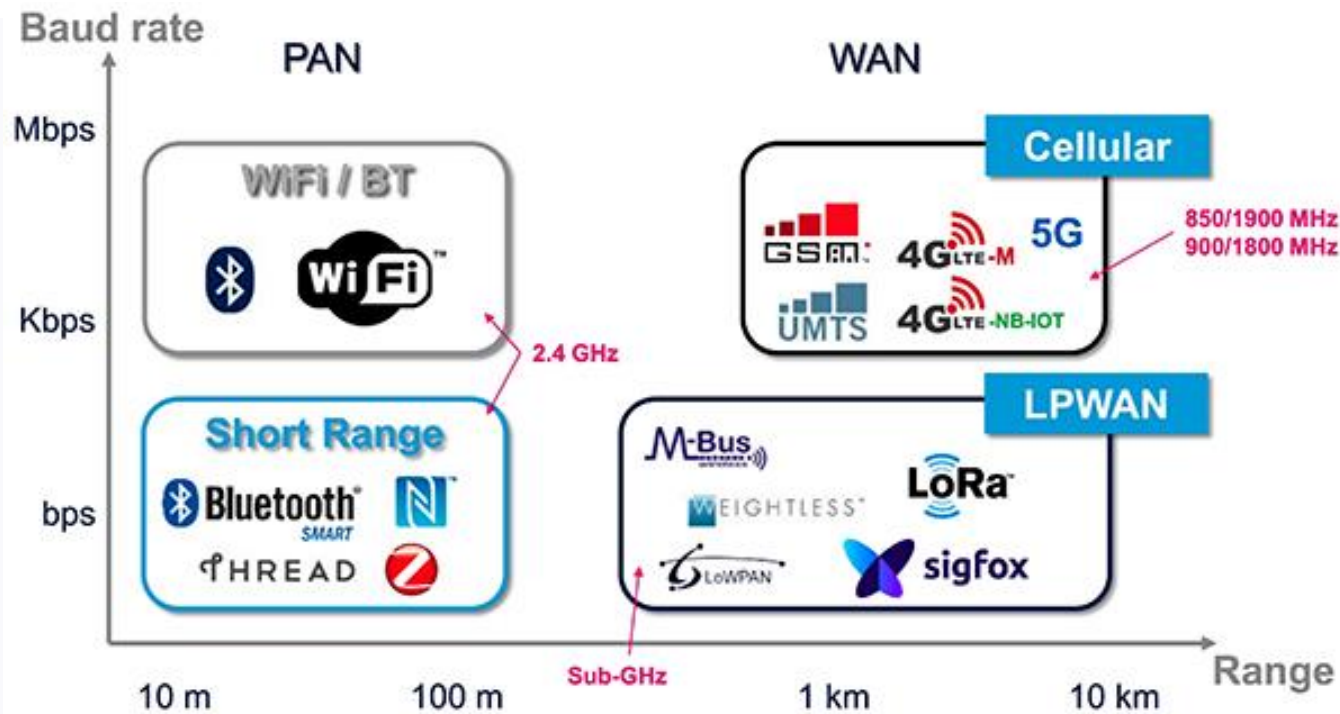


Image source: STMicroelectronics



Protocols – PHY and MAC Layers

Several protocols have been developed particularly for IoT, operating at layers 1 to 3 of the OSI model

IEEE802.15.4:

- Protocol specifying the physical and access layers for low-rate wireless personal area networks (LR-WPANs)
- Supports both sub-GHz and 2.4GHz
- Supports low-rate RF communication ranging from 20Kbps up to 250Kbps



- IEEE802.15.4-based specification for small, low-power digital radios
- 250Kbps data rate; 10-100 metres (line-of-sight) communication range
- A suite of several high-level protocols for Personal Area Networks
- Simpler and cheaper than other WPANs such as Bluetooth and Wi-Fi



Protocols – MAC and Networking Layers



- 
- IPv6 over Low power Wireless Personal Area Networks
 - One of the most important enabling techs for IoT
 - *Adaptation layer protocol* that enables IPv6 to run on 802.15.4 networks
- Defines the necessary encapsulation and header compression mechanisms
 - 6LoWPAN enables weak sensor motes to be assigned global IPv6 addresses
 - Developed by IETF 6lowpan Working Group ([RFC 6282](#))

IPv6

- Latest version of the Internet Protocol
- Provides the address space to address every single object in the world for the next many-many years.



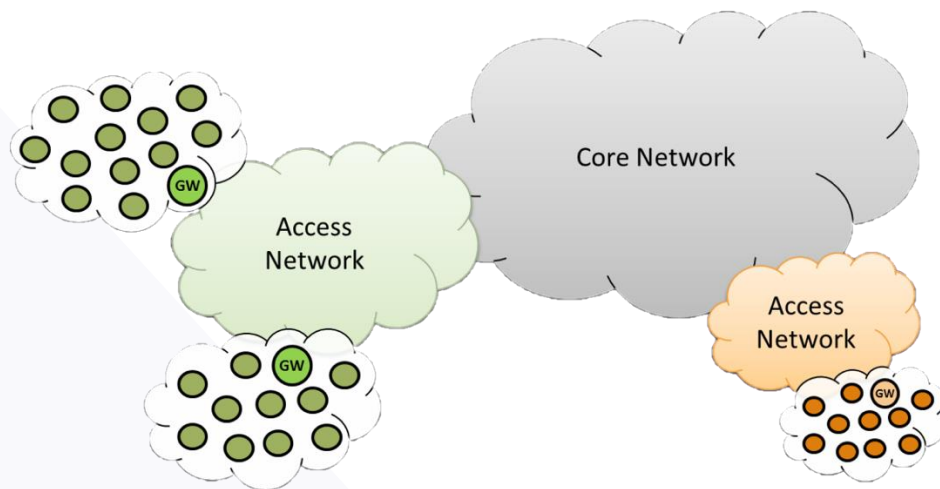
The RPL protocol

IP for Smart Objects

- Set of IPv6-based solutions being defined by IETF
- Supported by the IPSO alliance (<http://www.ipso-alliance.org/>)



I E T F®





IPv6 for Smart Objects

Interoperability

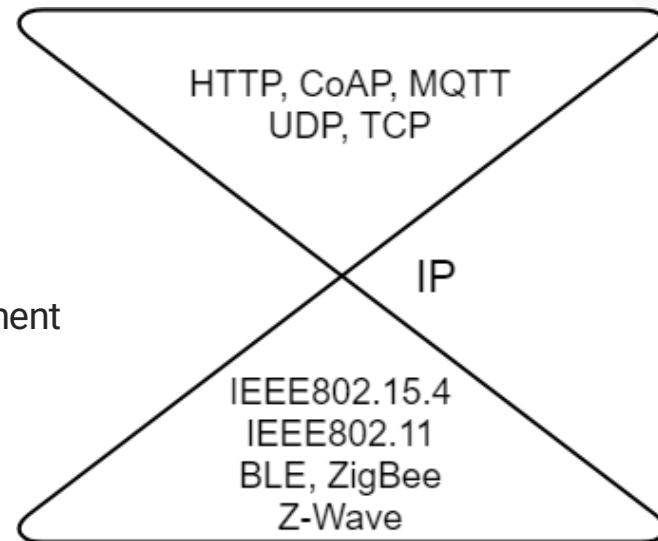
- layered approach for independence of underlying technologies

Scalability

- Survived the current Internet evolution
- unique global addressing
- Direct support for self-configuration and management

End-to-end

- No multi-protocol intermediate gateways that:
 - Are expensive and difficult to manage
 - Lack end-to-end QoS
 - Pose security issues





Routing Protocol for Low-power and Lossy Networks (RPL)

Terminology:

DAG: Directed Acyclic Graph. A directed graph having the property that all edges are oriented in such a way that no cycles exist. All edges are contained in paths oriented towards and terminating at one or more root nodes.

A DAG root is a node within the DAG that has no outgoing edge.

Rank: A scalar indicating the depth of a node (position in DAG)

RPLInstanceID: Is a unique identifier within a network.

Objective Function (OF): Defines how routing metrics, optimization objectives, and related functions are used to compute Rank.

Metrics:

Metrics:

- Node Residual Energy: $EE = \frac{Power_{now}}{Power_{max}} \cdot 100$

- ETX:

$$PRR(\rho) = \frac{\text{Number of received packets}}{\text{Number of sent packets}}$$

$$ETX = \frac{1}{PRR_{down} \cdot PRR_{up}}$$

DIO: DODAG Information Object

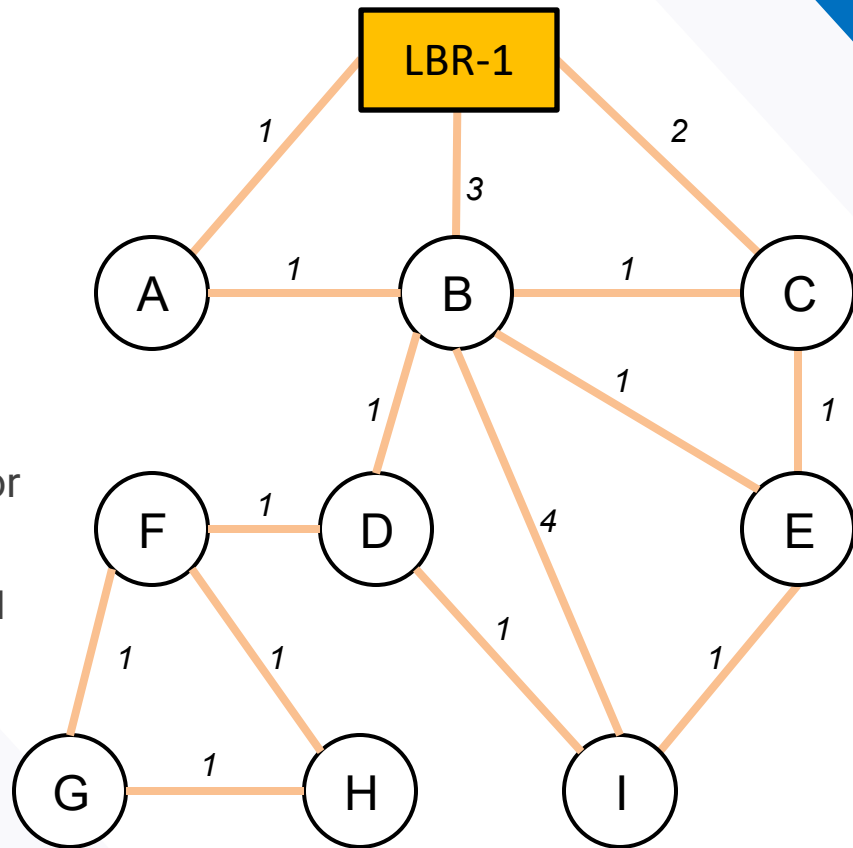
DAO: Destination Advertisement Object

DIS: DODAG Information Solicitation



DAG Construction

- LLN links are depicted
- Links are annotated w/ ETX
- It is expected that ETX variations will be averaged/filtered to be stable enough for route computation
 - Nodes are also to observe the metric and gain confidence before use

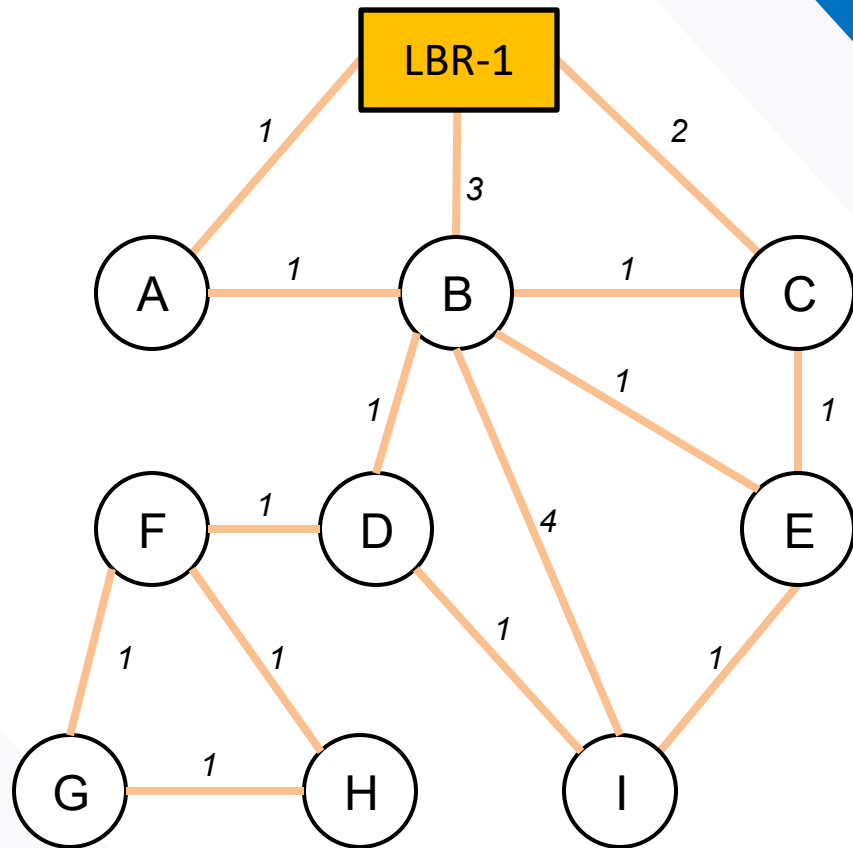




DAG Construction

Objective Code Point for example

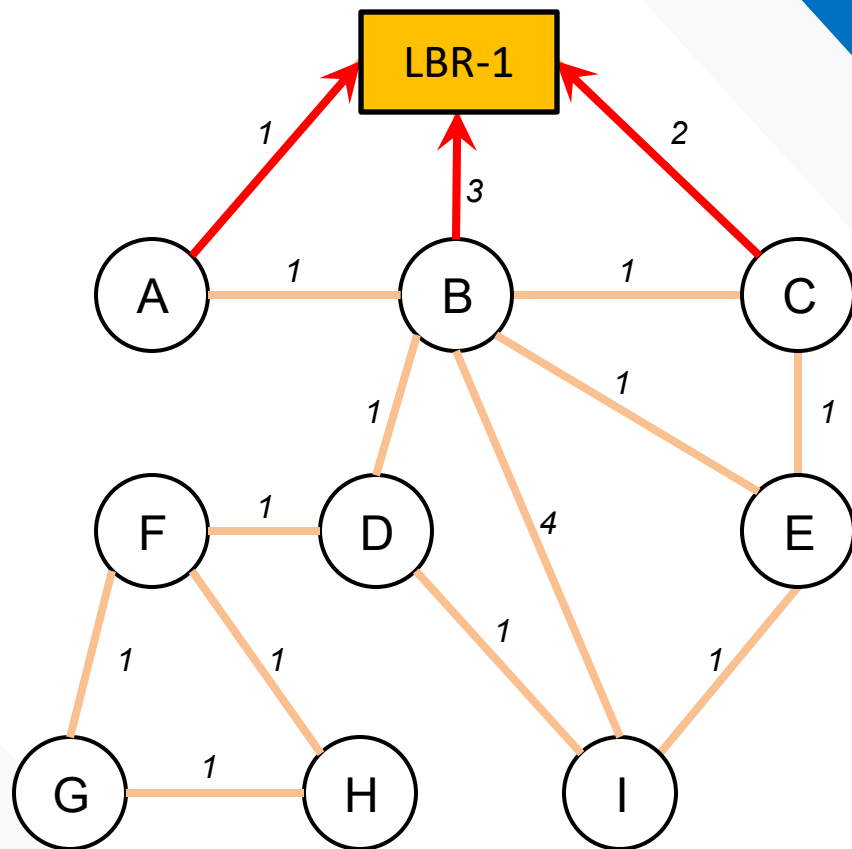
- Metric: ETX
- Objective: Minimize ETX
- Depth computation: Depth $\sim$ ETX
 - Note that a practical computation may be more coarse





DAG Construction

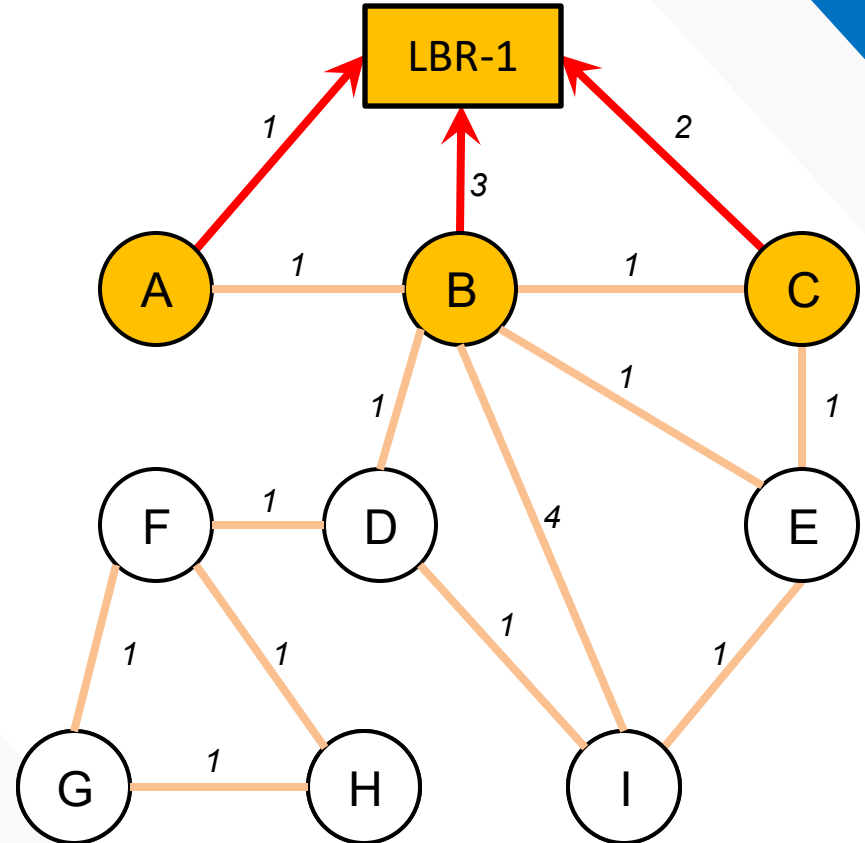
- LBR-1 multicasts RA-DIO
- Nodes A, B, C receive and process RA-DIO
- Nodes A, B, C consider link metrics to LBR-1 and the optimization objective
- The optimization objective can be satisfied by joining the DAG rooted at LBR-1
- Nodes A, B, C add LBR-1 as a DAG parent and join the DAG





DAG Construction

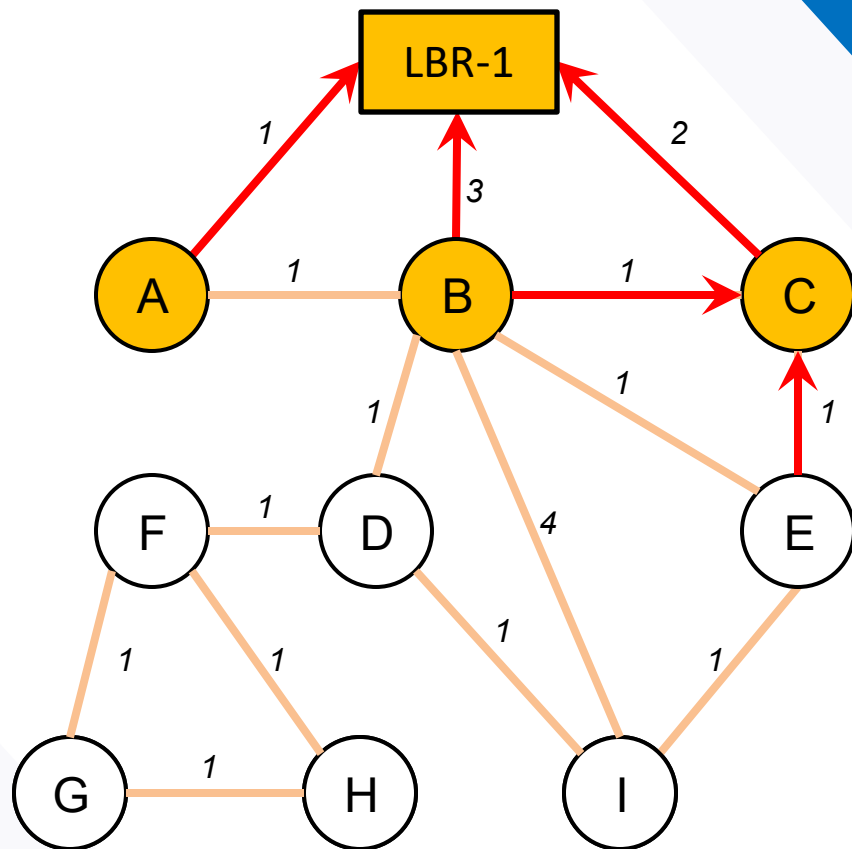
- Node A is at Depth 1 in the DAG, as calculated by the routine indicated by the example OCP (i.e., Depth ~ ETX)
- Node B is at Depth 3, Node C is at Depth 2
- Nodes A, B, C have installed default routes (::/0) with LBR-1 as successor





DAG Construction

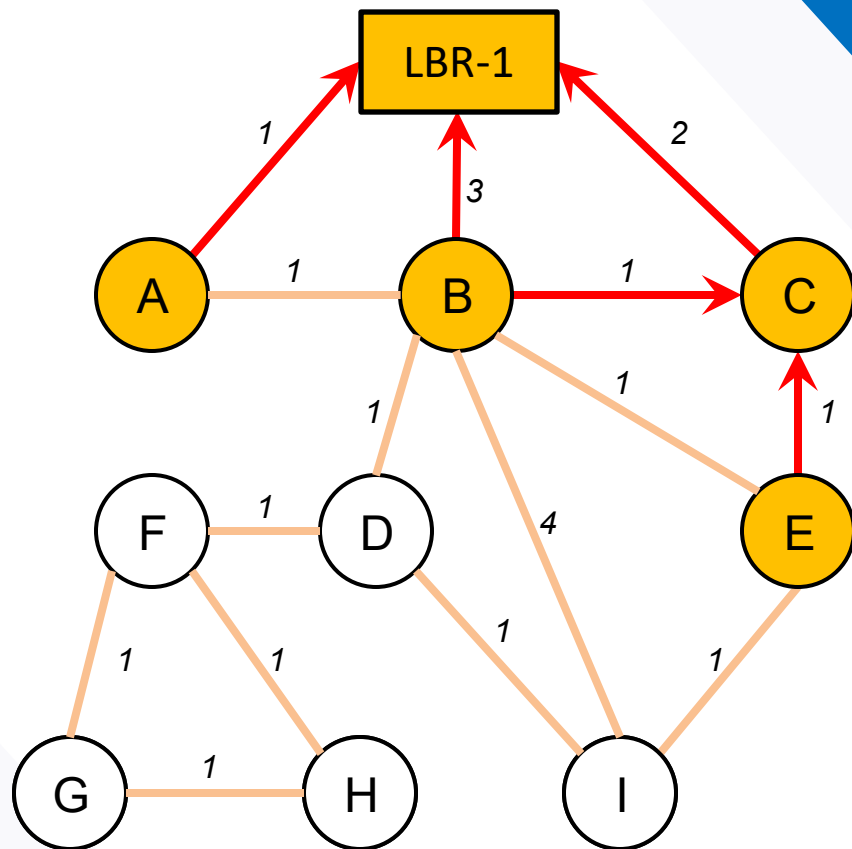
- The RA timer on Node C expires
- Node C multicasts RA-DIO
- LBR-1 ignores RA-DIO from deeper node
- Node B can add Node C as *alternative* DAG Parent, remaining at Depth 3
- Node E joins the DAG at Depth 3 by adding Node C as DAG Parent





DAG Construction

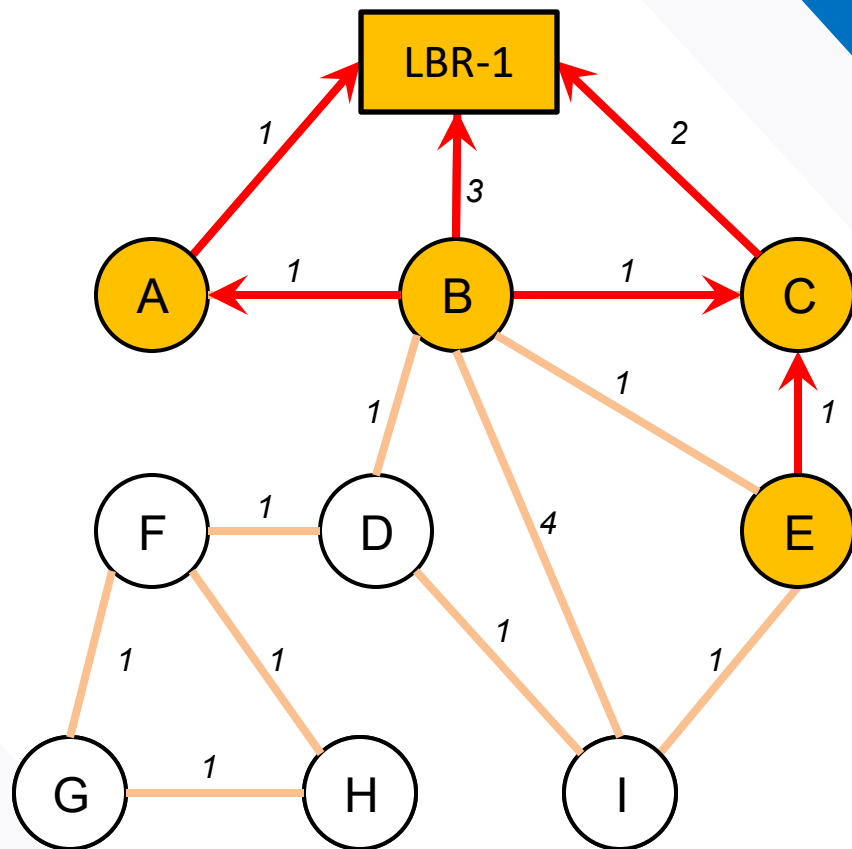
- The RA timer on Node C expires
- Node C multicasts RA-DIO
- LBR-1 ignores RA-DIO from deeper node
- Node B can add Node C as *alternative* DAG Parent, remaining at Depth 3
- Node E joins the DAG at Depth 3 by adding Node C as DAG Parent





DAG Construction

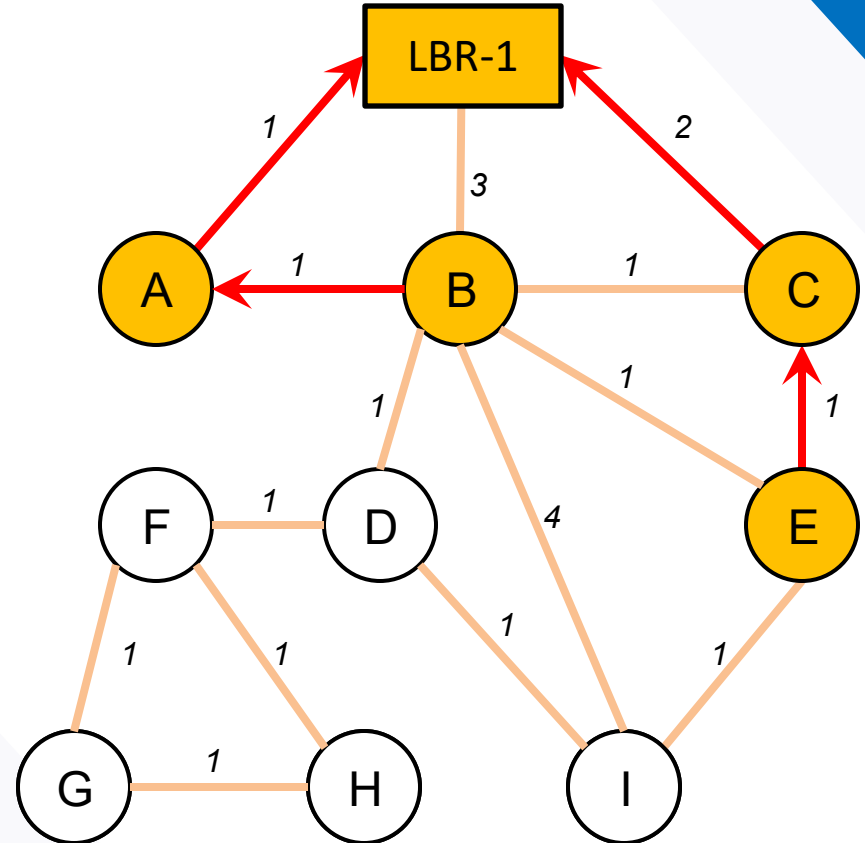
- The RA timer on Node A expires
- Node A multicasts RA-DIO
- LBR-1 ignores RA-DIO from deeper node
- Node B adds Node A
- Node B can move to a better position in the DAG
- Node B *removes* LBR-1 and Node C as DAG Parents





DAG Construction

- Node A is at Depth 1, $::/0$ via LBR-1 with ETX 2
- Node B is at Depth 2, $::/0$ via A with ETX 2
- Node C is at Depth 2, $::/0$ via LBR-1 with ETX 2
- Node E is at Depth 3, $::/0$ via C with ETX 3



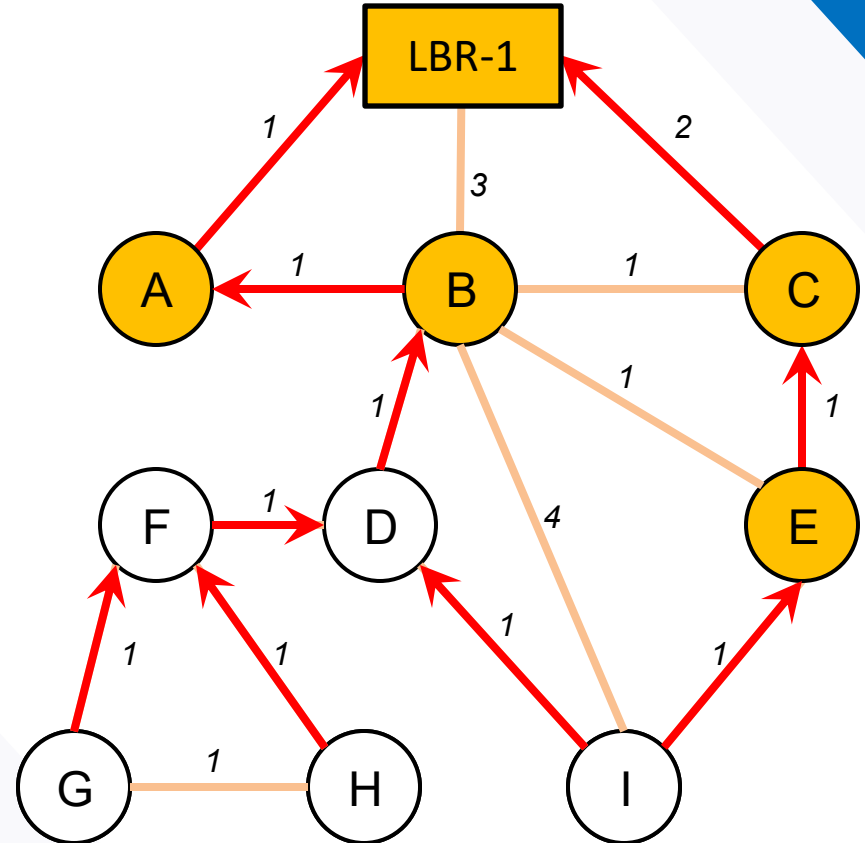


DAG Construction

- DAG Construction continues...
- And is continuously maintained

RPL also provides mechanisms for

- Multiple Points to Point communication (MP2P)
- Point to Multiple Points communication (P2MP)
- Loop removal





Low Power Wide Area Networks

Wireless communication technologies that balance the **trade-off** between **communication range** and **power consumption**.

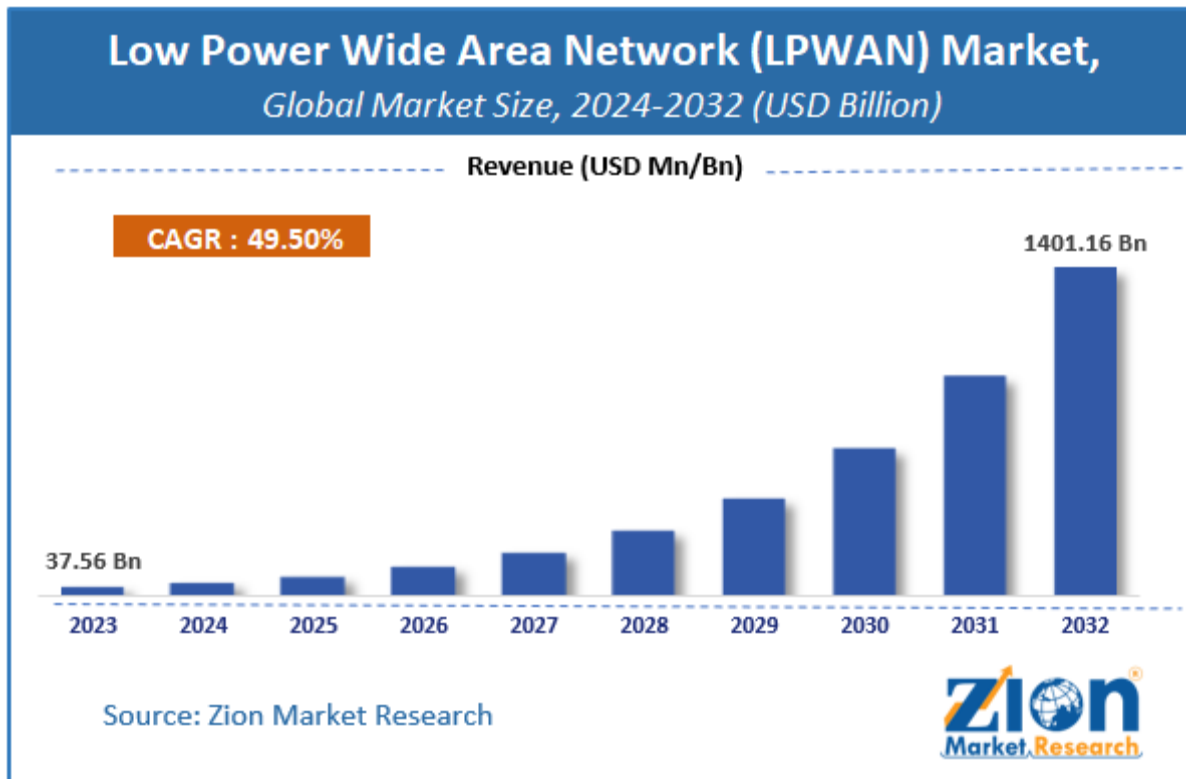
- Low frequency (sub-GHz) => long range communication links (Kms)
- Low bitrate => low duty cycle => low power consumption (few μAh per transmission)
- Typically, these are non-IP networks
- LPWAN operate in both licenced (e.g. NB-IoT) and unlicensed spectra (e.g. LoRaWAN, SigFox)
- There is no winning tech yet.

Choice depends on app requirements and coincidental conditions (e.g. area coverage by each tech)





LPWAN Market Size

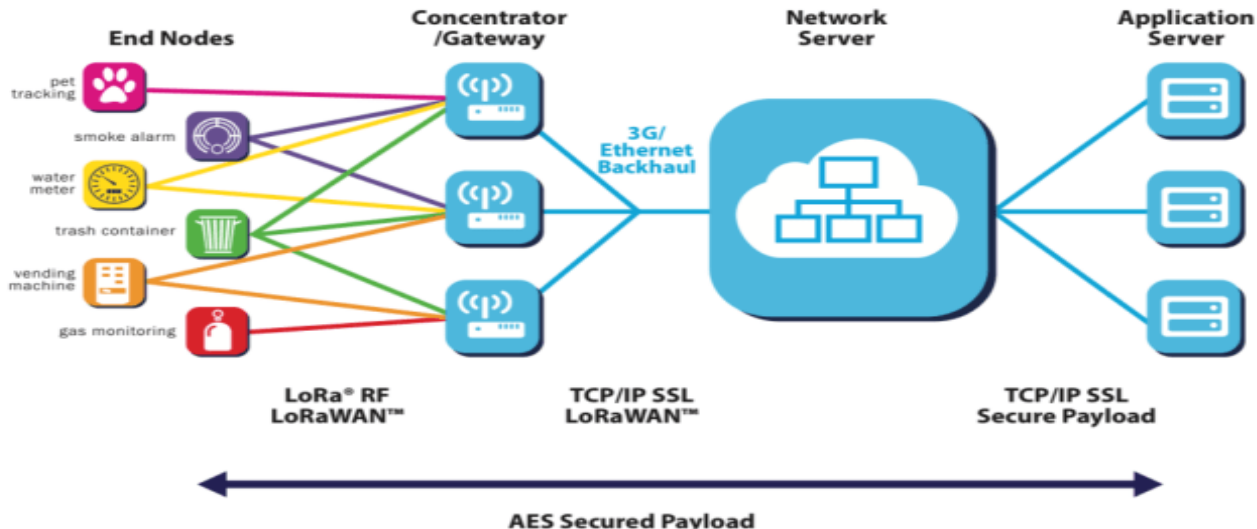




LoRaWAN

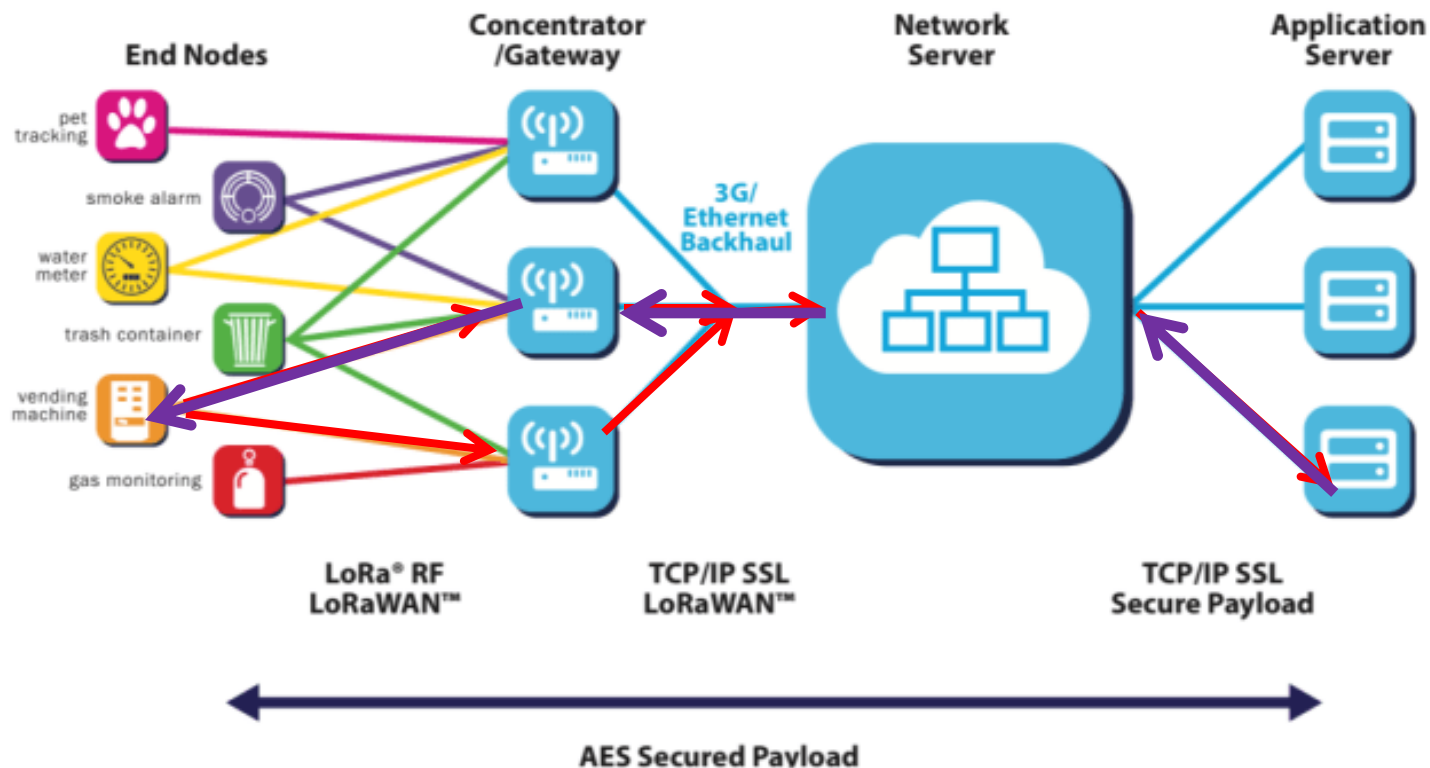


- A Wide Area Network protocol for IoT
- PHY & MAC layer protocol designed for large-scale public networks
- Data rates 0.3Kbps – 50Kbps; Communication range up to few Kms
- Architecture: Multicast; msgs from devices are received by multiple APs





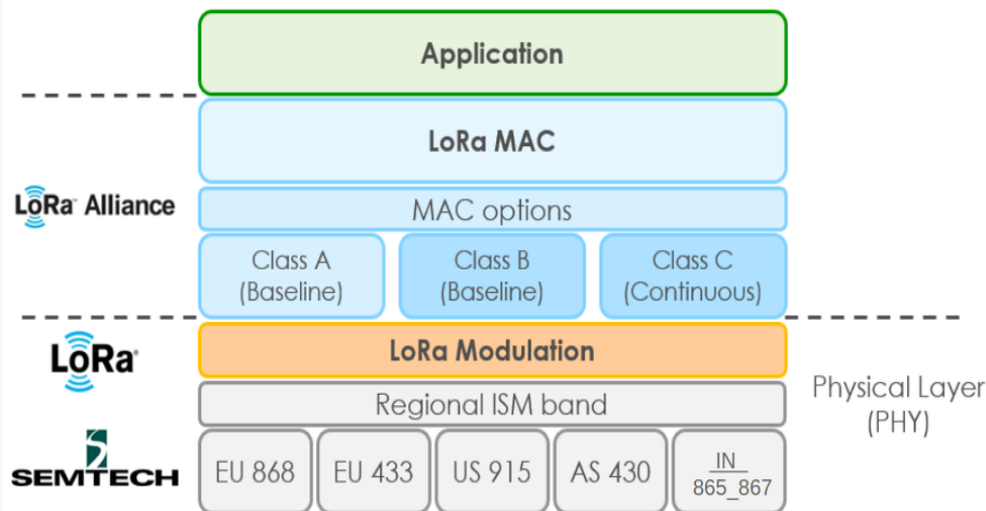
LoRaWAN Network Architecture





LoRaWAN Network Stack

LoRaWAN is the network which uses LoRa as the underlying radio modulation



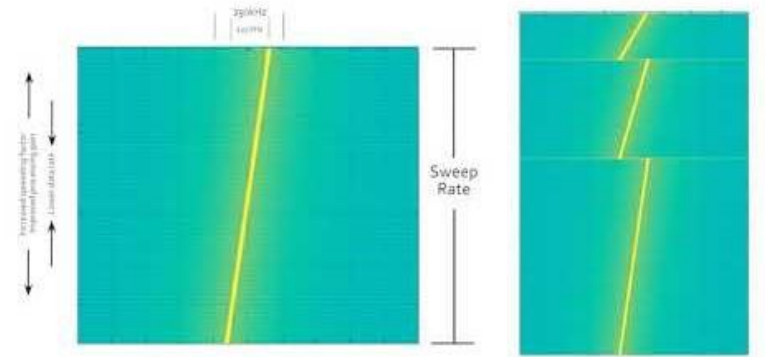
3 classes of devices

- A. Battery powered devices, bi-directional Comm (**low power consumption, high latency**).
- B. Battery powered devices, bi-directional comm., extra receive windows (**high power consumption, low latency**).
- C. Main powered devices (**Mains-powered devices, no latency**)



LoRaWAN – Signal Modulation

- **LoRa** (Long Range) is a proprietary modulation technique for wireless signals, owned by Semtech.
- An ALOHA-like protocol
- Spread spectrum technology
 - Chirp modulation (CSS)
 - Low power, long range at expense of low data rate
 - Resistance to interference
- Defines the physical layer: i.e. the wireless modulation utilised to create long distance communication link



LoRa Chirp Modulation

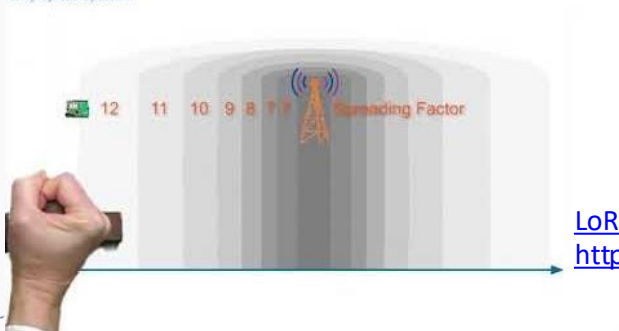
<https://www.youtube.com/watch?v=dxYY097ONs0>



LoRaWAN – Signal Modulation (contin'd)

- LoRaWAN operates in unlicensed frequencies, set via international agreements
 - In Europe 868MHz and 433MHz with 250KHz and 125KHz channels
 - A 1% duty cycle is imposed (e.g. if TimeOnAir = 0.5secs => wait time = 49.5secs)
- Spread Factor (SF): a key characteristic of LoRaWAN
 - It affects the bitrate and the fly-time (time to transmit) of a message
 - Time on Air can range from few msecs up to 30secs (<https://www.loratools.nl/#/airtime>)

Chirp Spread Spectrum



[LoRa Spreading Factor](https://www.youtube.com/watch?v=B580NvdXtjs)

<https://www.youtube.com/watch?v=B580NvdXtjs>



LoRaWAN – Spread Factor

The **Spread Factor** is a tunable parameter in LoRa modulation that trades **data rate for range and robustness**

It directly affects **range, data rate, time on air, power consumption, and robustness** of the communication.

The Spread Factor determines how many chips are used to encode each symbol in a LoRa transmission.

LoRa uses Chirp Spread Spectrum (CSS) modulation.

Each symbol is represented by $2^S F$ chips.

Typical SF values: SF7 to SF12

Symbol duration T_s :

$$T_s = \frac{2^{SF}}{BW}$$

For example, at SF12 and bandwidth 125 kHz:

$$T_s = \frac{4096}{125000} \approx 32.8 \text{ ms}$$

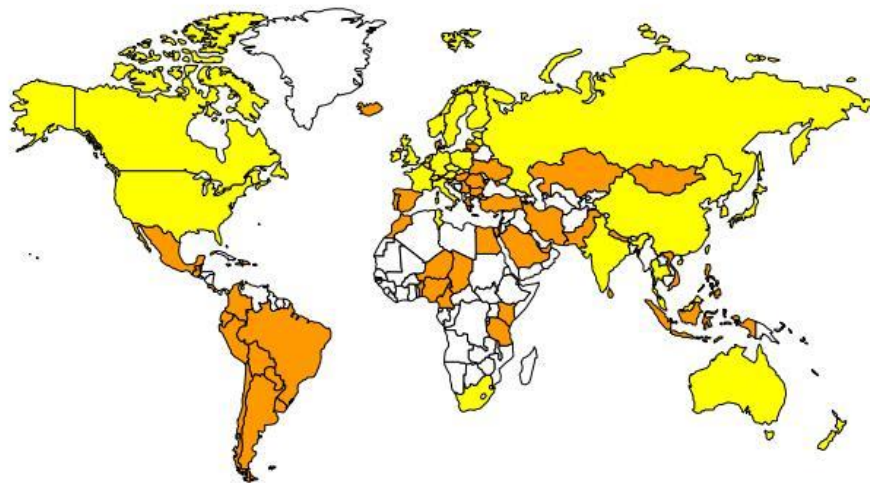
At SF7 and bandwidth 125 kHz :

$$T_s = \frac{128}{125000} \approx 1.02 \text{ ms}$$



LoRa Alliance Ecosystem

Countries — LoRaWAN™ Networks



LoRa Alliance™

June 2017

- 42 Publicly Announced Operators
- 30 Alliance Member Operators
- 250+ on-going trials & city deployments
- 480+ members in the Alliance

Legend:

- Publicly Announced
- Other Deployments



Reasons for Success

- While LoRa (the modulation technique) is proprietary, there is a plethora of commercially available products (APs and end-devices).
- One can deploy and maintain their private LoRaWAN network.
- However:



<https://www.thethingsnetwork.org/>

CATAPULT
Digital





Protocols – Networking Layer



- Also a Low Power Wide Area Network protocol for IoT
- However, it uses proprietary technology and requires proprietary infrastructure
- Ultra narrowband: sub-Ghz frequencies; 140 uplink msg/day and 4 downlink msg/day
- Architecture: 1-hop star topology between each AP and the devices





Protocols – Networking Layer



- Narrow-Band IoT
- Standard specifications developed by 3GPP (i.e. comes from the Mobile Network Operators)
- Focuses specifically on **indoor coverage**, low cost, long battery life, and high connection density.
- Uses a subset of the LTE standard; [OFDM](#) modulation for downlink communication and [SC-FDMA](#) for uplink communications.
- It will be delivered over the existing cellular infrastructure (in the UK pioneered by Vodafone)



Protocols – Transport Layer



CoAP
Constrained Application
Protocol

CoAP – Constrained Application Protocol

- Designed for use in resource-constrained devices, such as WSN nodes
- Brings RESTful architectures to IoT
- It's the HTTP of IoT - easily translated / interoperable with HTTP
- IoT resources can be abstracted as REST resources
- E.g. you can GET, POST and PUT a sensor or a device
- RESTful Machine2Machine communication over the Internet
- IETF Constrained RESTful Environments (CORE) [RFC 7959](#)

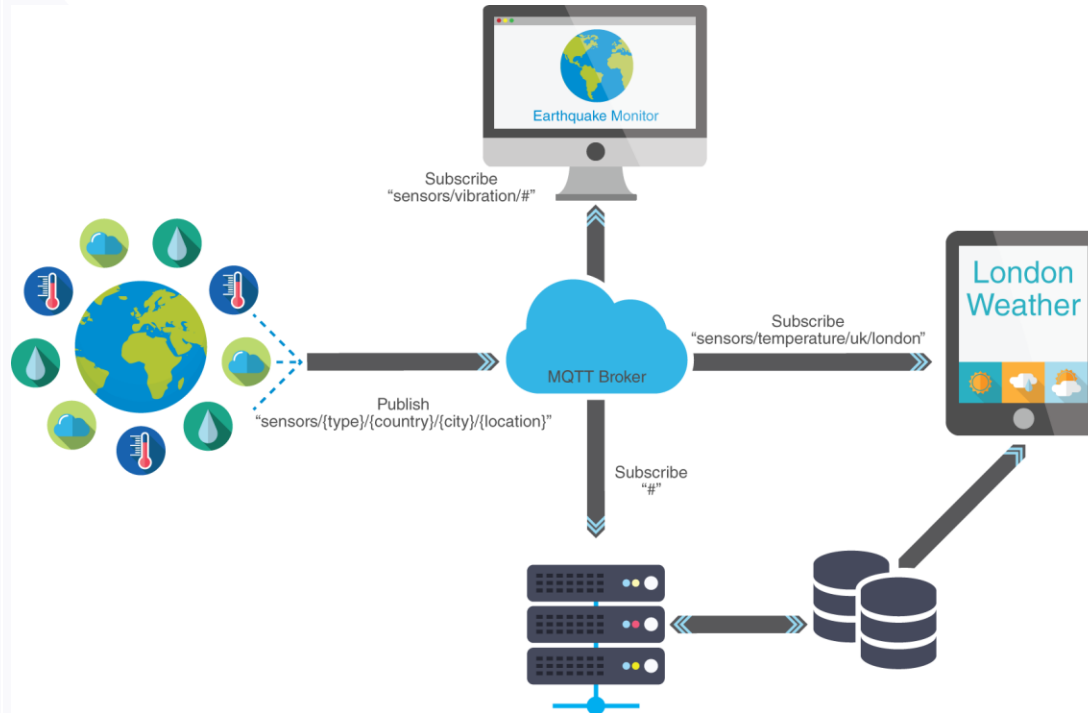


- 
- INTERNATIONAL
HELLENIC
UNIVERSITY



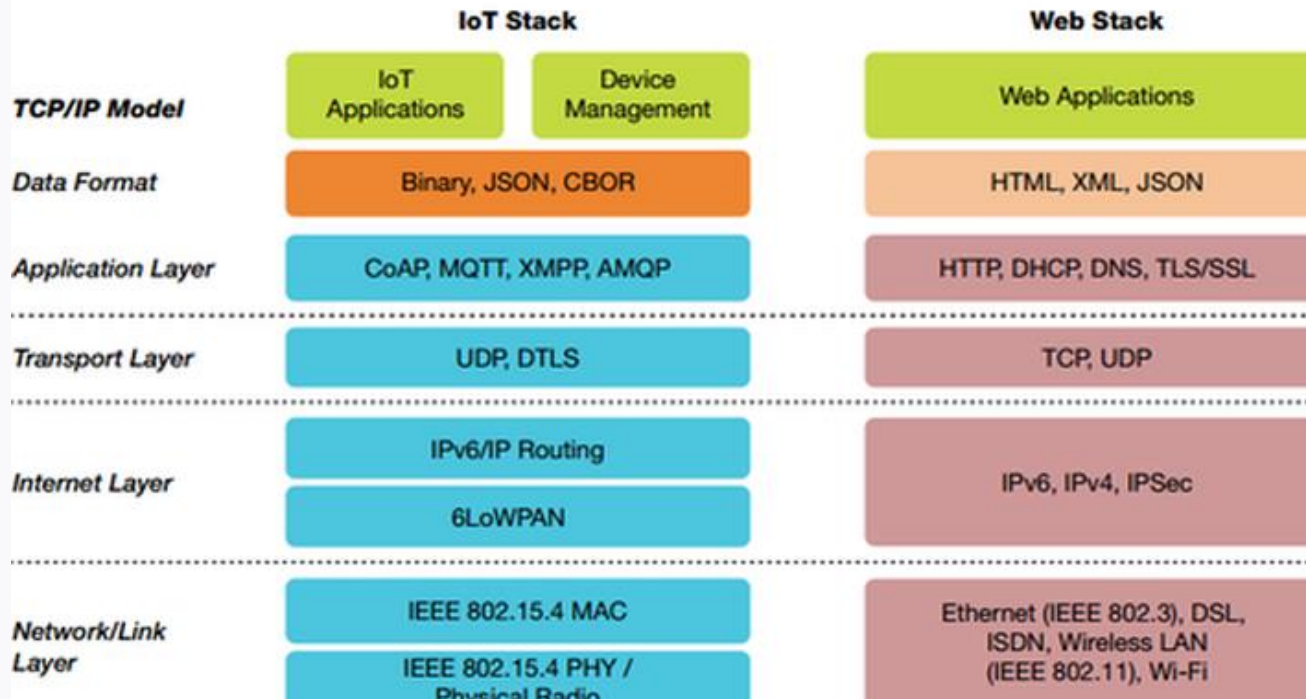
Protocols – Transport Layer

MQTT - Message Queue Telemetry Transport



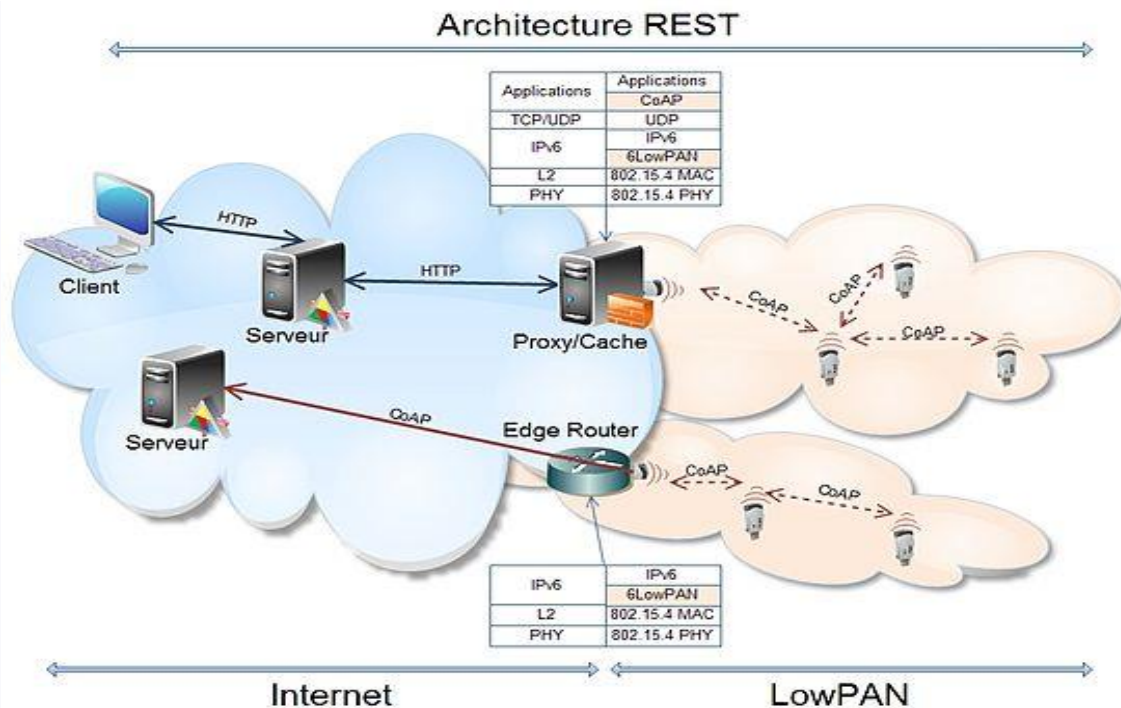


IoT Architectures – Net Stack



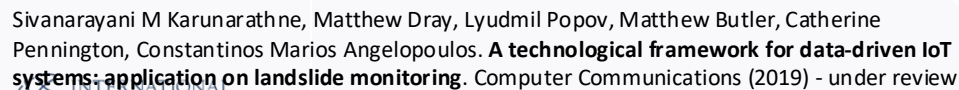
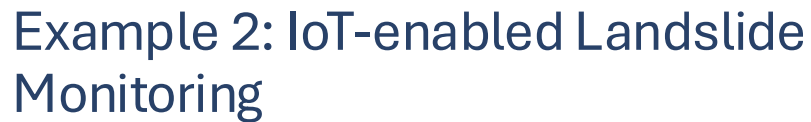


IoT Architectures – Example 1



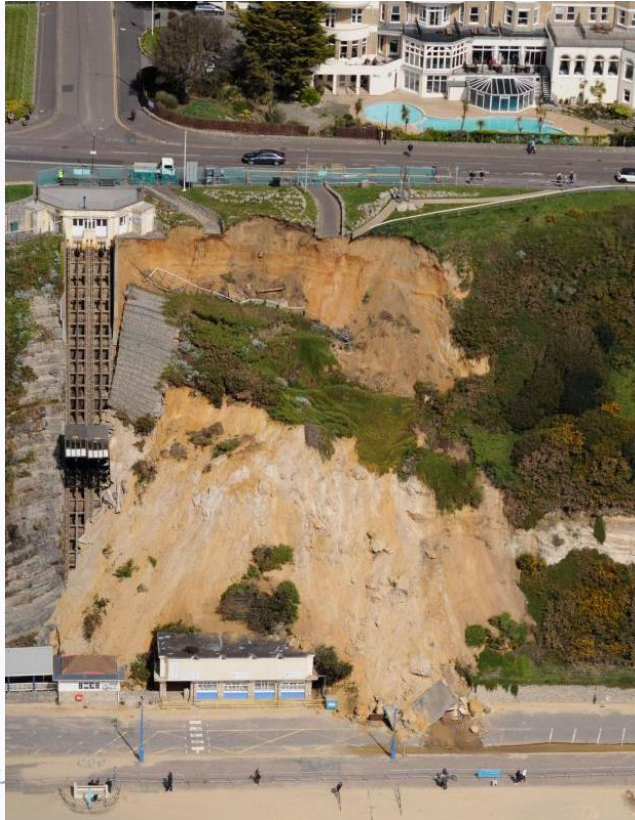
Application Layer
CoAP Methods
CoAP Transactions
UDP
Network Layer (IPv6)
6LoWPAN adaptation layer
802.15.4 MAC Layer
802.15.4 PHY Layer

Source: Wikipedia



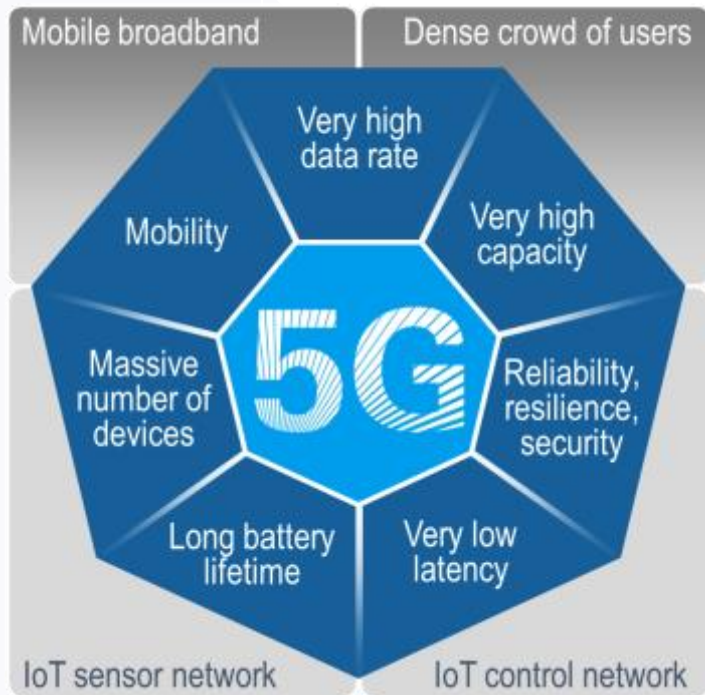


Example 2: IoT-enabled Landslide Monitoring



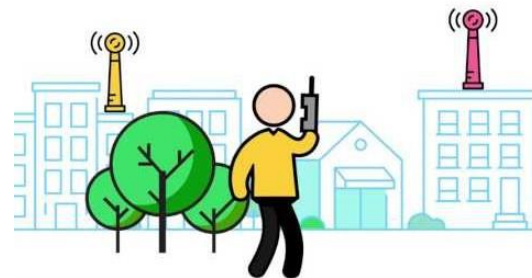


IoT and 5G



Video by IEEE Spectrum

https://www.youtube.com/watch?v=GEx_d0SjvS0

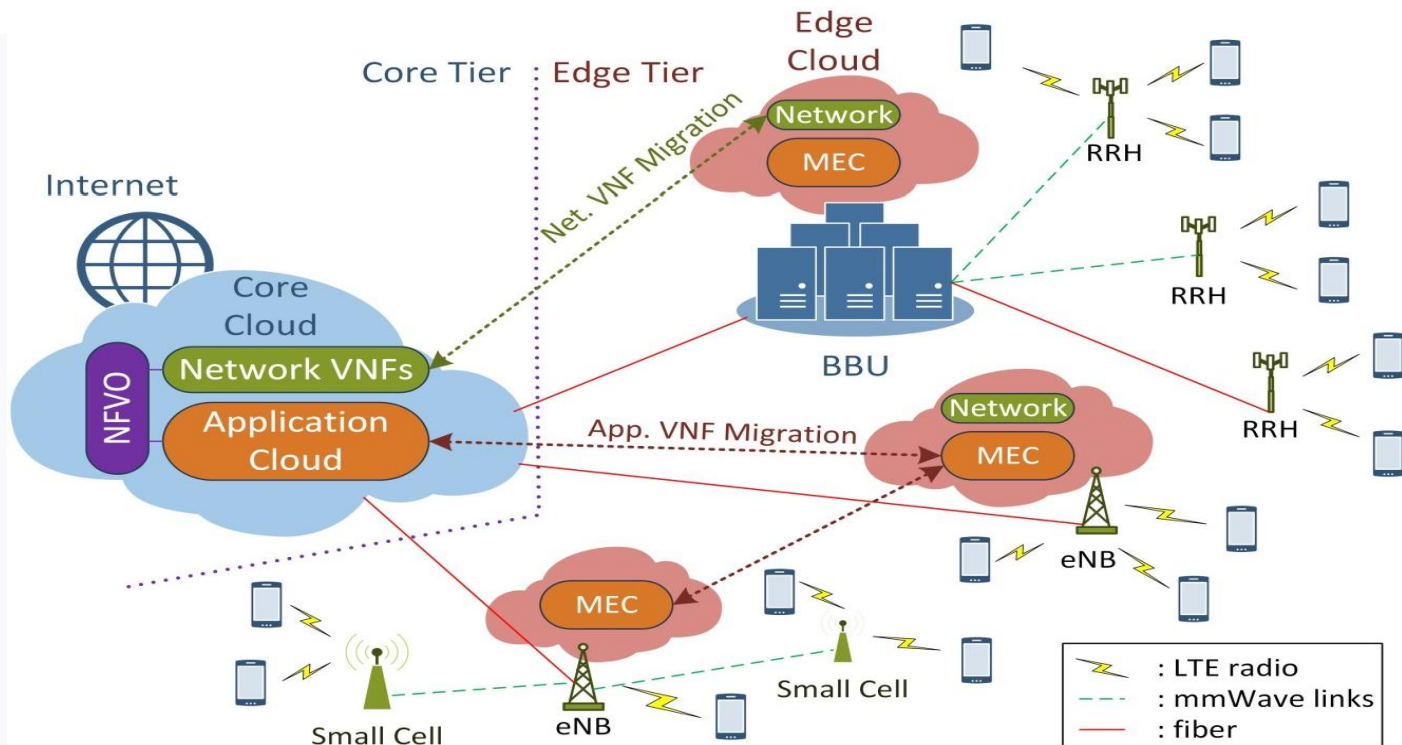


Quick Facts:

- 5G refers to an ecosystem of technologies
- It will employ also mmWave frequencies (24GHz, 60GHz)
- The focus is on **network services**; not network technologies
- It is yet to be standardised!



MEC-enabled 5G Architecture





- **NFV:** Network Function Virtualisation
- **NFVO:** Network Function Virtualisation Orchestrator
- **MEC:** Multi-access Edge Computing
- **BBU:** BaseBand Unit
- **RRH:** Remote Radio Heads
- **eNB:** eNodeB – LTE access network



Department of Computer Engineering
& Electronic Systems



INTERNATIONAL
HELLENIC
UNIVERSITY

Μέχρι την επόμενη φορά

Καλό σας απόγευμα

Prof. Marios Angelopoulos

mangelopoulos@ihu.gr

